Consent Guarantees for Multi-Step Assistance

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Abstract

A model can learn to anticipate needs and still overstep when permission is uncertain. We derive the maximal action set that satisfies a veto in every situation compatible with an observation. A finite audit checks 28,672 combinations of observations and permissions. For multi-step assistance, we characterize which permitted first moves retain a guaranteed route to completion or safe handoff. A legal preparation can fail this test when it creates a commitment that cannot be canceled. A separate synthetic training comparison distinguishes verified permission from imputed approval. An exact maximal-set representation can shrink the continuation check when several possible histories share the same available actions.

Summary and scope

Question. What may an assistant do when authority is partly observed and later steps could require separate consent?

Finding. The intersection of compatible permission sets governs each effect. A finite-horizon continuation test identifies which permitted first actions can still finish or hand off safely in every compatible state.

Evidence and scope. Exact finite checks cover immediate permissions and multi-step continuations. The fitted comparison uses synthetic supervision. The guarantees require valid permission sets and transitions that include every feasible authority change.

1 Preference and authority

Cooperative inverse reinforcement learning studies how a robot can infer a human's objective from interaction [1]. Inferring that someone probably wants an outcome leaves a separate question about who may choose how to pursue it. A development example should make that distinction explicit. Compare offering to reschedule a meeting with actually moving it. The latter also affects colleagues and may require authorization the current user cannot give.

Let $C(o)$ be the nonempty set of states still compatible with a trusted observation o. In state s, let $P(s)$ contain the actions permitted by all relevant authorities under the declared policy. Include separate actions for asking, deferring, and reversible assistance. The set $P(s)$ can vary with jurisdiction and action scope.

Proposition 1. *The maximal set of actions safe against any compatible veto is*

$$P_*(o) = \bigcap_{s \in C(o)} P(s).$$

A randomized action policy guarantees this property exactly when its support lies in $P_*(o)$.

Proof. Each member of the intersection lies in every permission set. Each excluded action has at least one compatible state in which it is forbidden. Assigning that action positive probability gives positive probability of violating a veto in that state. Conversely all mass inside the intersection is permitted in each state. □

Place estimated benefit inside this permission boundary. Even with correct gains, a positive weighted sum can compensate for a protected person's loss by assigning sufficiently large gains to others. The companion finite check enumerates 125 three-person gain vectors. Of these, 27 have a positive sum and at least one negative component. A protected welfare floor would reject those vectors. Actual authorization is a separate input to $P(s)$, and a predicted gain cannot provide it.

2 Permission across a sequence of effects

Suppose permitted preparatory actions can change the available later choices. Let $T(s, a)$ be the nonempty set of possible successor states after executing action a in state s. Let the resulting trusted observation z reveal a subset of those successors. Include authorization changes and any effects bundled with the action in this model. A set G consists of states in which the workflow may finish or hand off safely. Stopping in a state outside G fails the stated completion requirement.

For a current compatible set B, define $V_0(B)$ to hold exactly when $B \subseteq G$. An action a is viable with h steps remaining if $h > 0$, it is permitted throughout B, and for each observation z with a nonempty successor set

$$B_{a,z} = \{s' \mid s' \in T(s, a) \text{ for some } s \in B, \text{ the observation is } z\}$$

we have $V_{h-1}(B_{a,z})$. Set $V_h(B)$ to hold when $B \subseteq G$ or there is at least one viable action. This is a finite-horizon use of the state-set construction for games with imperfect information [2].

Proposition 2. *The viable actions are exactly the first actions of policies that obey all permissions and finish or hand off within h steps in every modeled history. Under a pathwise guarantee, a randomized policy can use only such first actions.*

Proof. At horizon zero the only decision is to stop, which is safe exactly when all compatible states are terminal. At positive horizon the first action must satisfy the immediate permission intersection. Every reachable observation leaves a continuation problem with one fewer step. A legal policy needs a continuation in each branch. Conversely, assembling one legal continuation per branch after a permitted action gives a legal policy. Induction proves the equivalence. A positive probability of a nonviable action gives positive probability to a choice without the required pathwise guarantee. □

Consider a document assistant that may either ask for consent or prepare a release in an external tool. Asking reveals whether it may commit, and a refusal permits safe handoff. Preparing is currently permitted in both worlds but creates a reservation. If consent is absent and the reservation has no permitted cancel path, preparation can strand the workflow. With a two-step horizon, only asking is viable. Adding a permitted cancel action makes preparation viable too. The cancel path must be enforced by the external tool, with a fresh permission check for every later effect.

For 128 four-state fixtures, an independent oracle enumerated all bounded policy trees and evaluated each in every compatible initial world and successor. It agreed with the recurrence on 512 model and horizon combinations. Across the positive horizons there were 341 immediately permitted first-action occurrences with no viable continuation. The counts describe the synthetic generator and do not estimate actual overreach.

3 Checking viability with maximal compatible sets

Viability is downward closed. A policy that works from B also works from any $B' \subseteq B$. Therefore it is enough to retain the inclusion-maximal viable sets $\mathcal{M}_h$. The initial family is $\mathcal{M}_0 = \{G\}$. For a given action a, let S_a be its set of permitted states. For observation z and a viable set W at horizon $h-1$, define

$$D_{a,z}(W) = \{s : \ T(s,a) \cap O_z \subseteq W\},$$

where O_z contains the states compatible with observation z. For the deterministic observation map used in the checker, these sets partition the state space.

Proposition 3. *The inclusion-maximal beliefs permitting first action a are the maximal members of the family*

$$\left\{ S_a \cap \bigcap_z D_{a,z}(W_z) : W_z \in \mathcal{M}_{h-1} \text{ for every } z \right\}.$$

To obtain $\mathcal{M}_h$, take the maximal members of the union of these action-specific families and $\{G\}$. The intersections can be computed one observation at a time, removing contained sets after each step.

Proof. An observation successor is viable exactly when it is contained in some maximal viable set at the preceding horizon. That holds for the successor after a and z precisely when every initial state lies in the corresponding $D_{a,z}(W_z)$. Intersect with S_a for immediate permission. An observation with no reachable successor imposes no additional constraint. This proves the family formula. If $P \subseteq Q$ are two partial intersections, then $P \cap D \subseteq Q \cap D$ for any later factor D. Dropping P therefore preserves the downward closure of the result. Stopping contributes G separately. □

Keep the maximal sets separately for each first action if the gate must return every viable choice. A system can cache these sets for a fixed permission model and look up a new belief by containment. A changed permission or transition rule requires rebuilding the relevant frontiers. This is an exact representation, with no general bound on the number of maximal sets that a model can produce.

A structured fixture used sixteen possible states, eight probe observations and four commit choices. Every state admitted the probe, which revealed its pair. The correct commit then depended on that pair. At the second horizon, direct expansion would enumerate 327,680 combinations across actions. The folded calculation used 160 pairwise intersections. This compares enumeration work on that fixture. The check also compared all first actions against a subset oracle across 131,072 belief and horizon pairs on the fixture and 9,216 on smaller models. One negative control sent two successor states to the same observation. Each admitted a different terminal move, and the gate rejected the shared predecessor because the operator could not know which move was permitted.

4 Labeling cases during training

Construct matched triples with the same helpful objective: explicit authorization, explicit veto, and missing or ambiguous authority. Label the minimal legitimate act in each case. Preserve examples where scope differs, such as permission to draft a message without permission to send it. Train a permission prediction head with evidence about who can authorize which effects. Use a structured loss on probability mass assigned to excluded actions and a separate loss rewarding practical, authorized help. A low average loss leaves individual violations possible, so action execution still needs a trusted, fresh authority check.

A permission query is valuable when it can distinguish the compatible states and a permitted route remains open. Some requests are intrusive or repetitive. Include cases where a reversible offer is a better choice than asking again. A refusal carries no negative engagement penalty in this curriculum.

For a multi-step curriculum, label each viable action from a declared transition model. Keep tests where asking reveals a refusal, authority expires between steps, or a previously offered cancel path fails. If a transition is possible but missing from the model, the guarantee is invalid. Treat approval evidence and its freshness as inputs to the effect gate.

5 Finite check and limits

Three latent states, four actions, and every nonempty compatible-state set give $16^3(2^3-1) = 28{,}672$ checks. Bitwise intersection agreed with a separately computed per-action veto scan. In the minimal witness, an action permitted by one compatible state and vetoed by another is excluded even if the permitted state is assigned high probability. The guarantee requires the true state to be represented and the permission sets to be correct. A fabricated authorization or omitted affected party defeats those premises. A later study should measure overreach and needless refusal separately, including revocation and cross-person authority.

6 Learning from imputed and verified permission

Each seed trained two logistic policies with explicit approval, veto, and unknown encoded separately. One training target treated 85% of unknown cases as approved. The other required observed affirmative authority. An independently sampled 3,200-case panel contained 788 authorized cases and 2,412 cases with a veto or unknown authority. The imputed target led to 1,641 unauthorized executions. The verified target led to zero. Both executed in all 788 authorized cases. This binary experiment tests the supervision rule for execution. It leaves the choice among asking, waiting, and giving safe assistance to a larger action study.

Review status. Independent manuscript review and a language-model study remain open.

References

[1] D. Hadfield-Menell, A. Dragan, P. Abbeel and S. Russell. Cooperative Inverse Reinforcement Learning. `https://arxiv.org/abs/1606.03137`.

[2] Krishnendu Chatterjee, Laurent Doyen, Thomas A. Henzinger, and Jean-François Raskin. *Algorithms for Omega-Regular Games with Imperfect Information*. Logical Methods in Computer Science 3(3:4), 2007. `https://arxiv.org/pdf/0706.2619`.